

AI Agent Assignment

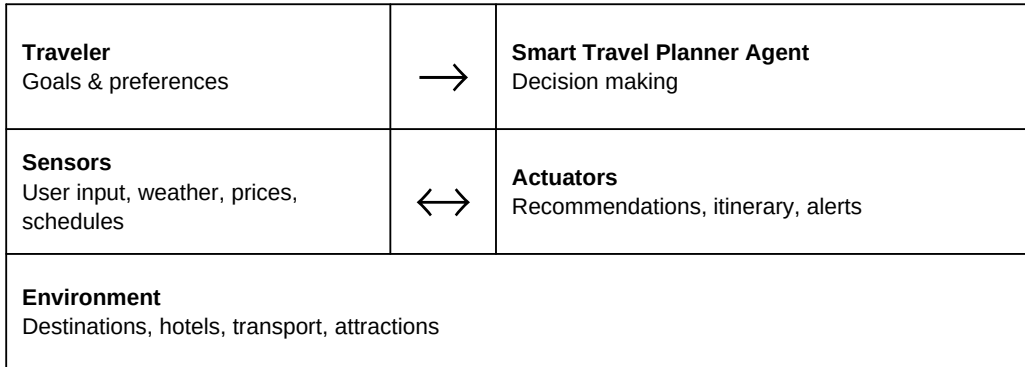
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Agent Name: Smart Travel Planner Agent

Purpose: An AI agent that helps travelers plan trips according to their destination, budget, preferences, transportation, weather and activities.

1. Graphical Representation



The traveler gives requirements. The agent senses information, processes it, selects a suitable plan and presents recommendations.

2. Agent Formulation

Component	Description
Agent	Smart Travel Planner Agent
Environment	Travel destinations, roads, airports, hotels, attractions, weather and changing prices.
Sensors	User input, weather, transportation schedules, prices and destination information.
Actuators	Destination recommendations, itineraries, route/activity suggestions and alerts.
Goal	Create an affordable, convenient and useful travel plan.

3. Environment Types

Type	Evaluation
Partially Observable	Yes — future weather, traffic, prices and preferences may be unknown.
Stochastic	Yes — weather, traffic, prices and delays can change unexpectedly.
Sequential	Yes — one travel decision affects later decisions.
Dynamic	Yes — travel conditions can change during planning.
Discrete	Mostly — destinations, hotels and activities are selectable options.
Multi-agent	Yes — travelers, airlines, hotels and service providers affect the environment.

4. PEAS Framework Evaluation

PEAS	Smart Travel Planner Agent
Performance Measure	Traveler satisfaction, budget compliance, suitable destination, reasonable travel time, safety and convenience.
Environment	Destinations, transport, hotels, attractions, weather, traffic and prices.
Actuators	Recommendations, itinerary, routes, activities and alerts.
Sensors	Traveler preferences, budget, dates, weather, schedules, prices and destination data.

5. Example Working

For example, a traveler enters Karachi, a budget of PKR 30,000, three days and an interest in beaches. The agent can recommend suitable activities and create a simple itinerary.

Conclusion

The Smart Travel Planner Agent senses information, makes decisions according to a goal, and acts by providing useful travel recommendations. The PEAS framework clearly defines its performance, environment, actuators and sensors.